

Python Lists & Loops

Fun with Coding for Grade 3!

What Are Lists in Python?

A **list** is a wonderful way to store many things together in one single place!

 Think of it like a **basket holding different fruits**, or a box holding all your favorite toys!

```
fruits = ["apple", "banana",  
"mango"]
```

Why Do We Use Lists?

- ★ **Keep Info Together:** Group connected items easily.
- ★ **Easy to Find:** Simple to pick and look at any item.
- ★ **Smart Memory:** Helps our computer program remember many facts at once!

How to Make a List

Use **square brackets []** to create your list!

Remember these rules:

- Put items inside the brackets.
- Separate each item with a **comma (,)**.
- Wrap text words inside quotes.

Meet Our Friends!

Let's connect code to real life with lists of our friends and foods:

```
friends = ["Asha", "Ravi",  
"Meena", "Arjun"]  
  
favorite_foods = ["samosa",  
"idli", "jalebi", "paneer"]
```

What Is a Loop?

Loops help us do the exact same action many times automatically!

 It's just like repeating a fun dance move—you do the same steps again and again for every beat!

The 'for' Loop in Python

We use a for loop to look at **each item** inside our list, one after the other!

```
for fruit in fruits:  
    print(fruit)
```

What Happens Inside a Loop?

Each time the loop runs, Python performs three steps:

1. **Picks** the next item from the list.
2. **Does the action** (like running the `print()` command).
3. **Moves forward** automatically to the next item until done!

Looping Through Our Friends!

```
for friend in friends:  
    print(friend)
```

Output:

```
Asha  
Ravi  
Meena  
Arjun
```

Let's Print Favorite Foods!

```
for food in favorite_foods:  
    print(food)
```

Output: samosa, idli, jalebi, paneer

Using Loops to Say Hello!

We can add text inside our loops to give everyone a custom greeting!

```
for friend in friends:  
    print("Hello, " + friend +  
        "!!")
```

Output:

```
Hello, Asha!  
Hello, Ravi!  
Hello, Meena!  
Hello, Arjun!
```

Common Mistakes to Avoid!

- **Missing Brackets:** Always wrap your lists with square brackets [].
- **Forgetting the Colon:** Don't forget the colon : at the end of your for line!
- **Spaces Matter (Indentation):** Always indent/space out the print line inside your loop!

What Did We Learn Today?



-  Lists hold many items together in one variable.
-  We use [] to create lists in Python.
-  Loops repeat actions for every item instantly.
-  Coding is full of creativity and fun!

Try It Yourself!

Create a list of your 3 favorite animals and print each one using a loop!

```
animals = ["cat", "dog",  
           "elephant"]  
  
for animal in animals:  
    print(animal)
```

★ You can do it! ★

More Fun to Explore!

- Add more items to your lists!
- Try using numbers or favorite colors.
- Combine lists to tell your own stories in code!

★ Great job! Keep exploring and keep coding! ★